

Group 2

Arham Lodha

September 12, 2023

Claim 1a. *Let $q : G \rightarrow G'$ be a surjective homomorphism. Prove that if G is cyclic, then G' is cyclic.*

Proof. Let $q : G \rightarrow G'$ be a surjective homomorphism. Suppose G is a cyclic group. Then by definition of cyclic groups, there exists a $g \in G$ s.t $G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\}$. Since, q is surjective, it means for all $x \in G'$, $\exists y \in G$ s.t $q(y) = x$. By definition of a homomorphism, $q(g^k) = \prod_{i=1}^k q(g) = q(g)^k$ for all $k \in \mathbb{Z}$. Thus every element in G' can be written as $q(g)^k$ for some $k \in \mathbb{Z}$. Thus let $h = q(g)$, since every element in G' can be written as $h^k = q(g)^k$ for some $k \in \mathbb{Z}$, then $G' = \{h^k : k \in \mathbb{Z}\}$. Thus every element in G' can be generated by h . Thus by definition of cyclic groups, G' is cyclic. $\square$

Claim 1b. *Let $q : G \rightarrow G'$ be a surjective homomorphism. If G is abelian, then G' is abelian.*

Proof. Let $q : G \rightarrow G'$ be a surjective homomorphism. Suppose G is abelian. By definition of abelian groups, $\forall a, b \in G$ $ab = ba$. Thus $q(ab) = q(ba)$. By definition of homomorphisms, $q(ab) = q(a)q(b)$ and $q(ba) = q(b)q(a)$. Thus $q(a)q(b) = q(b)q(a)$. Since q is surjective, $\forall x, y \in G'$, there exists a $a, b \in G$ s.t $q(a) = x$ and $q(b) = y$. Now since $q(a)q(b) = q(b)q(a)$ for all $a, b \in G$, then $xy = yx$. Thus G' is commutative and thus is abelian. $\square$

Claim 2a. *The intersection $K \cap H$ of subgroups of a group G is a subgroup of H .*

Proof. Suppose K , and H are subgroups of G . By definition of a intersection, $K \cap H \subseteq H$.

1. **Identity:** $e \in G$ by the identity axiom of groups and H and K have a identity element by the same. Since the identity element is unique in a group and the identity element of any subgroup is the identity element of the parent group, $e \in K$ and $e \in H$. Thus $e \in K \cap H$.
2. **Inverse:** For any $a \in K \cap H$, since K and H are subgroups then by the inverse axiom $a^{-1} \in K$ and $a^{-1} \in H$. Thus $a^{-1} \in K \cap H$.
3. **Closure:** $\forall a, b \in K \cap H$, $ab \in K$ and $ab \in H$ by the closure property of groups. Thus $ab \in K \cap H$.

Thus $K \cap H$ is a subgroup of H . $\square$

Claim 2b. *If K is a normal subgroup of G , then $K \cap H$ is a normal subgroup of H .*

Proof. Suppose K is a normal subgroup of G . By definition of a intersection, $\forall a \in K \cap H$, $a \in K$ and $a \in H$. By definition of a subgroup, $\forall h \in H$, $h \in G$. By definition of a normal subgroups, $hah^{-1} \in K$. Furthermore since H is closed under the binary operation, $hah^{-1} \in H$. Thus $hah^{-1} \in K \cap H \forall a \in K \cap H$ and $\forall h \in H$. Thus $K \cap H$ is a normal subgroup. $\square$

Claim 3a. Let U denote the group of invertible upper triangular 2×2 matrices $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, and $\phi : U \rightarrow \mathbb{R}^\times$ be a map which sends $A \rightarrow a^2$. Prove that ϕ is a homomorphism.

Proof. Let U denote the group of invertible upper triangular 2×2 matrices $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, and $\phi : U \rightarrow \mathbb{R}^\times$ be a map which sends $A \rightarrow a^2$. For all $A, B \in U$. A and B can be written by definition of U as the following $A = \begin{bmatrix} a_1 & b_1 \\ 0 & c_1 \end{bmatrix}$ and $B = \begin{bmatrix} a_2 & b_2 \\ 0 & c_2 \end{bmatrix} \forall a_1, b_1, c_1, a_2, b_2, c_2 \in \mathbb{R}$. $AB = \begin{bmatrix} a_1 a_2 & a_1 b_2 + b_1 c_2 \\ 0 & b_1 b_2 + c_1 c_2 \end{bmatrix}$. $\phi(AB) = a_1 a_2$. $\phi(A) = a_1$ and $\phi(B) = a_2$. Thus for all $A, B \in U$, $\phi(AB) = a_1 a_2 = \phi(A)\phi(B)$. Thus by definition of a homomorphism, ϕ is a homomorphism. $\square$

Problem 3b. Determine ϕ 's kernel and image.

Solution. By definition of kernel of a homomorphism, its all the elements in the domain which map to 1. By definition of ϕ , which maps any invertible upper triangular matrix to the square of its top left corner element, its all invertible upper triangular matrices with ± 1 on the top left corner. Thus any matrix of the form $\begin{bmatrix} \pm 1 & b \\ 0 & c \end{bmatrix} \forall b, c \in \mathbb{R} \text{ s.t } c \neq 0$. The $\phi_*(U) = \mathbb{R}$, because for each $a \in \mathbb{R}$ there exists a matrix $A = \begin{bmatrix} \sqrt{a} & 0 \\ 0 & 1 \end{bmatrix} \in U$ s.t $\phi(A) = a$. Thus the image of ϕ is $\mathbb{R}_+$ (Positive reals) and ϕ is surjective. $\blacksquare$

Claim 4a. Let $f : \mathbb{R}^+ \rightarrow \mathbb{C}^\times$ be the map $f(x) = e^{ix}$. f is a homomorphism.

Proof. Let $f : \mathbb{R}^+ \rightarrow \mathbb{C}^\times$ be the map $f(x) = e^{ix}$. For all $a, b \in \mathbb{R}^+$, $f(a) = e^{ia}$ and $f(b) = e^{ib}$ and $f(a)f(b) = e^{ia}e^{ib} = e^{i(a+b)}$ (properties of exponents). $f(a+b) = e^{i(a+b)} = f(a)f(b)$. Thus by definition of a homomorphism, f is a homomorphism. $\square$

Problem 4b. Determine f 's kernel and image.

Solution. By definition of kernel of a homomorphism, its all the elements in the domain which map to 1. By definition of f , $1 = e^{i2k\pi}$ for all $k \in \mathbb{Z}$. Thus by definition of f , $\ker f = \{2k\pi : k \in \mathbb{Z}\}$. The image of f , is any element on the unit circle in the complex plane since complex exponentiation is the same as rotation in the complex plane. $\blacksquare$

Claim 5a. Let H be the set of $n \times n$ matrices that have the block form $M = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}$ with $A \in GL_r(\mathbb{R})$ and D in $GL_{n-r}(\mathbb{R})$, form a subgroup H of $GL_n(\mathbb{R})$.

Proof. Let H be the set of $n \times n$ matrices that have the block form $M = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}$ with $A \in GL_r(\mathbb{R})$ and D in $GL_{n-r}(\mathbb{R})$.

For all $X \in H$, $X = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}$ for $A \in GL_r(\mathbb{R})$ and $B \in GL_{(n-r)}(\mathbb{R})$. $\det(X) = \det(A)\det(D)$, since $\det(A) \neq 0$ and $\det(D) \neq 0$ because A and D are invertible. Thus $\det(X) \neq 0$. Thus X is invertible. $H \subset GL_n(\mathbb{R})$.

1. **Identity:** $I_n \in H$, since $A = I_r$, $B = 0$, and $D = I_{n-r}$. Thus H has the identity.
2. **Inverse:** By definition since $H \subset GL_n(\mathbb{R})$, then each element H is invertible thus there exists a $N \in GL_n(\mathbb{R})$. $N = \begin{bmatrix} A^{-1} & -A^{-1}BD^{-1} \\ 0 & D^{-1} \end{bmatrix}$. $MN = \begin{bmatrix} AA^{-1} & -AA^{-1}BD^{-1} + BD^{-1} \\ 0 & DD^{-1} \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & I_{n-r} = I_n \end{bmatrix}$. $NM = \begin{bmatrix} AA^{-1} & A^{-1}B - A^{-1}BDD^{-1} \\ 0 & DD^{-1} \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & I_{n-r} \end{bmatrix} = I_n$. Thus every M has a N inverse.
3. **Closure:** For all $X, Y \in H$. $X = \begin{bmatrix} A_1 & B_1 \\ 0 & C_1 \end{bmatrix}$ and $Y = \begin{bmatrix} A_2 & B_2 \\ 0 & C_2 \end{bmatrix}$. $XY = \begin{bmatrix} A_1 & A_1B_2 + C_1C_2 \\ 0 & C_1C_2 \end{bmatrix} \in H$. Thus H is closed.

Thus H is a subgroup of $GL_n(\mathbb{R})$. □

Claim 5b. The map $H \rightarrow GL_r(\mathbb{R})$ that sends $M \rightarrow A$ is a homomorphism.

Proof. Let $f : H \rightarrow GL_r(\mathbb{R})$ be the map that sends $M \rightarrow A$. For all $X, Y \in H$. $X = \begin{bmatrix} A_1 & B_1 \\ 0 & C_1 \end{bmatrix}$ and $Y = \begin{bmatrix} A_2 & B_2 \\ 0 & C_2 \end{bmatrix}$. $XY = \begin{bmatrix} A_1 & A_1B_2 + C_1C_2 \\ 0 & C_1C_2 \end{bmatrix}$. $f(X) = A_1$ and $f(Y) = A_2$. $f(X)f(Y) = A_1A_2$. $f(XY) = A_1A_2$. Thus $f(XY) = f(X)f(Y)$. Thus by definition of homomorphism, f is homomorphism. □

Problem 5c. What is the kernel of the map?

Solution. The kernel of f is all the matrices in H of the form $M = \begin{bmatrix} I_r & B \\ 0 & D \end{bmatrix}$ for all $D \in GL_{n-r}(\mathbb{R})$. Thus $f(M) = I_r$. ■

Problem 6. Determine the center of $GL_n(\mathbb{R})$.

Claim 6a. Any matrix in the center commutes with a off diagonal matrix

Proof. Let $e_{i,j}(k)$ be the matrix of 0s with $a_{i,j} = k$. $e_{i,j}(k)$ is a off diagonal matrix. Every matrix in the center commutes with every element of $GL_n(\mathbb{R})$ by definition. Thus any matrix in the center commutes with a off diagonal matrix $e_{i,j}(k)$ □

Claim 6b. Any matrix which commutes with a off diagonal matrix is diagonal.

Proof. Assume the contrary, A is not a diagonal matrix, s.t exists a $a_{i,j} \neq 0$ where $i \neq j$ which commutes with $B = e_{i,j}(1)$. Left multiplying by B is the same as taking the j th row and making it the i th row while putting a 0 everywhere else. Right multiplying by B is the same as taking the i th column and making it the j th column. Thus $a_{j,j} \neq 0$ of AB while $a_{j,j} = 0$ of BA . Thus $BA \neq AB$. Thus by contradiction, any matrix commutes with a off diagonal matrix has no off diagonal elements thus it is diagonal. □

Thus all elements of the center are diagonal.

Claim 6c. Any diagonal matrix which commutes with all permutation matrices is scalar

Proof. Assume the contrary, A is a diagonal matrix which is not scalar but commutes with all permutation matrices. Thus there exists i and j s.t $a_{ii} \neq a_{jj}$. Let $P_{i,j}$ be the permutation matrix which when right multiplied swaps the i th and j th columns. However when left multiplying, $P_{i,j}$ swaps the i th and j th rows. $(ab)_{i,j} = a_{i,i}$ of AB while $(ba)_{i,j} = a_{j,j}$ of BA . Thus $AB \neq BA$. Thus by contradiction A must be scalar. $\square$

Thus all elements of the center must be scalar matrices.